

COMBINATORICS

ASSIGNMENT 1

ROB TOMPKINS

1. There are 100 students at a school and three dormitories A , B , and C with capacities 25, 35 and 40 respectively

- (a) How many ways are there to fill the dormitories?
- (b) Suppose that of the 100 students 50 are men and 50 are women and that A is an all-men's dorm, B is an all-women's dorm, and C is co-ed. How many ways are there to fill the dormitories.

Solution. Problem 1

□

2. A committee of five people is to be chosen from a club that boasts a membership of 10 men and 12 women. How many ways can the committee be formed if it is to contain at least two women? How many ways if, in addition, one particular man and one particular woman who are members of the club refuse to serve together on the committee?

Solution. Problem 2

□

3. A secretary works in a building located nine blocks east and eight blocks north of his home. Every day he walks 17 blocks to work.

- (a) How many different routes are possible for him?
- (b) How many different routes are possible if the one block in the easterly, which begins four blocks east and three blocks north of his home is water (and he can't swim)? (Hint: Count the routes that use the block under water).

Solution. Problem 3

□

4. There are 20 identical sticks lined up in a row occupying 20 distinct places as follows:
||||| - six of them are to be chosen.

- (a) How many choices are there?
- (b) How many choices are there if no two of the chosen sticks can be consecutive?
- (c) How many choices are there if there must be at least two sticks between each pair of chosen sticks?

Solution. Problem 4

□

Date: January 14, 2026.

5. How many integral solutions of $x_1 + x_2 + x_3 + x_4 = 30$ satisfy $x_1 \geq 2$, $x_2 \geq 0$, $x_3 \geq -5$, and $x_4 \geq 8$?

Solution. Problem 5

□

6. A bagel store sells six different kinds of bagels. Suppose you choose 15 bagels at random. What is the probability that your choice contains at least one bagel of each kind? If one of the kinds of bagels is Sesame, what is the probability that your choice contains at least three Sesame bagels?

Solution. Problem 6

□

7. Four (standard) dice (cubes with 1, 2, 3, 4, 5, 6, respectively, dots on their six faces), each of a different color, are tossed, each landing with one of its faces up, thereby showing a number of dots. Determine the following probabilities:

- (a) The probability that the total number of dots shown is 6
- (b) The probability that at most two of the dice show exactly one dot
- (c) The probability that each die shows at least two dots
- (d) The probability that the four numbers of dots shown are all different
- (e) The probability that there are exactly two different numbers of dots shown

Solution. Problem 7

□